Human Recognizer for Industrial Robots

A Lightweight Machine Vision Application for Safety during Human-Robot Interaction

ISE 572 Final Project

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Section 1: Problem Statement

As more factories adopt collaborative robots, there is a growing need for safer and more natural ways for humans and robots to work together. Many robot systems still rely on buttons, control panels, or voice commands for interaction. In a real factory, that can be a problem. It may be noisy, workers may be wearing gloves, and stopping to use a control panel can slow the task down.

This project looks at a machine vision approach for human-robot cooperation in an industrial setting. The main goal is to create a vision system that can detect when a human is present near a robot. That is the baseline objective. If that works well, the next goal is to identify the person's hands. After that, as a stretch goal, the system will attempt to recognize a small set of hand or body gestures that could be used as simple robot commands.

This fits well within Industry 4.0 because smart factories depend on flexible automation, real-time sensing, and safer human-machine interaction. A robot that can visually recognize a nearby worker could respond more appropriately and safely than one that only reacts to fixed sensors or manual input. In the future, this kind of system could support simple visual communication between workers and robotic platforms in environments where voice commands are unreliable.

There may also be value beyond traditional manufacturing. Similar problems exist in military and defense environments where robots may need to operate near humans in loud, communication-limited, or fast-moving settings. A robot that can visually detect a nearby person, notice raised hands, or eventually recognize basic signals could be useful in maintenance bays, logistics areas, expeditionary workspaces, or other human-machine teaming situations where speech is difficult and quick visual understanding matters.

If this problem is not addressed, there are several costs. Safety is the biggest one. A robot that cannot reliably detect a nearby person may require stricter barriers or slower operation. There is also a productivity cost when workers have to stop what they are doing to use a screen or button interface. On top of that, traditional safety and presence-detection systems may require multiple dedicated sensors, which can increase cost and complexity.

This project applies mainly to manufacturing and industrial automation, especially in settings where collaborative robots share space with human workers. It also has possible relevance to defense support and military human-machine teaming environments where local, reliable, and low-latency vision systems could improve coordination. The prototype developed in this project is intended to be portable, meaning the machine vision capability could later be transferred to many different robotic platforms that need this kind of awareness.

Section 2: Data Considerations

The data for this project would be video data collected from a camera observing a robot work area. The system would need examples of humans working on the scene. To balance the technical depth and feasibility of our project, we plan to use a pretrained model for detection and segmentation and train our own model for the classification step. To successfully train the classification model, we would need to collect video footage with the different poses signaling different commands or no command at all.

After we set up the pretrained detection and segmentation model, we would begin collecting video footage where the hands are clearly visible. That would allow the system to attempt hand identification. Once we are able to apply the pretrained model to our collected footage, the dataset would need to be labeled by gesture commands, such as stop, start, slow, or ready. These gestures would need to be chosen carefully so they are visually distinct and realistic for a factory environment.

For a classroom prototype, this data could come from a webcam or camera mounted near a robot or work area. We could record short video clips of a few people standing, moving into frame, raising their hands, and making simple gestures. Frames from those videos could then be labeled and used for training and testing. In a real industrial system, the data would likely come from fixed cameras or robot-mounted cameras installed around collaborative work cells.

There are a few data challenges that stand out. Lighting is one of the biggest. Factory lighting can create shadows, glare, and uneven brightness. Another challenge is variation between people. Different body sizes, clothing, gloves, and movement styles can make consistent detection and segmentation harder. Then when gesture recognition is attempted, the problem becomes even more sensitive to camera angle, occlusion, and motion blur.

Another issue is data volume. Hand and gesture recognition will likely need many examples to work reliably. Labeling the data correctly also matters and may be tedious. Similar hand positions could be mistaken for different gestures if the labels are not consistent.

To improve robustness, the dataset could be expanded with basic augmentation such as changes in brightness, slight rotation, zoom, or horizontal shifts. That would help the model handle more real-world variation without needing an enormous amount of original footage.

Section 3: Proposed ML Solution

This project is primarily a computer vision problem. The first task is to use a pretrained model to detect whether a human is present in the robot's nearby workspace. Once that stage is successful, the next task is to locate or identify hands via segmentation. The final goal is to classify a small number of gestures that could be mapped to simple robot commands.

So the project has a staged structure. Step one is human detection. Step two is hand-focused detection. Step three is gesture classification. This phased approach keeps the project realistic while still allowing room for extension if the early work goes well. The final model output after stage three is to classify a gesture such as stop, start, slow, or ready and use these commands appropriately at the robot.

For our pretrained model, we would likely use a vision transformer since this is the state of the art for detection and segmentation. For our classification model we would use convolutional neural networks, since CNNs are the best at cheap image-based tasks. If we find that our pretrained model is lacking in some specific way relative to our task, we are also open to fine-tuning, but this is not planned.

For stage two where hand and gesture segmentation would be attempted, pose-estimation or landmark-based methods could also help. Those methods can identify key body or hand points and may make gesture classification easier.

Long term, the idea is for this vision system to run locally on an edge device such as an Intel NUC. That matters because factory systems may not always want to rely on cloud processing for speed, privacy, or reliability reasons. The same edge-based approach could also be useful in military or field settings where connectivity is limited or unavailable. The result would be a prototype machine vision capability that could later be adapted to different robot platforms that need local visual awareness.

Section 4: Expected Impact

The main benefit of this project is that it could help robots become more aware of the humans around them. Even a basic human-detection system would be useful in collaborative industrial settings. A robot that can recognize when a person enters its workspace could slow down, stop, or switch behavior based on that information. That could improve both safety and workflow.

When the project reaches the later stages, hand and gesture recognition will make the interaction even more useful. A worker might eventually be able to give a simple visual signal without needing to press a button or rely on voice commands in a loud environment. That would be especially valuable in factories where noise and protective equipment make normal communication harder.

This project also supports Industry 4.0 goals by combining automation, sensing, and machine learning into a flexible smart-factory capability. Instead of building a solution around one specific robot, the intent is to create a prototype vision system that could be moved to other robotic platforms with similar needs.

There is also a possible secondary benefit for military support environments. Robots used around maintainers, operators, or logistics personnel may face many of the same communication challenges seen in industrial spaces. A vision system that can reliably notice people and interpret simple visual cues could support safer human-machine teaming in those settings as well, especially when local processing is important.

Success could be measured in several ways. For the baseline system, the most important metrics would be human detection accuracy, false positive rate, and response time. In other words, how often the system correctly recognizes a person, how often it mistakenly detects one, and how quickly it reacts. Once hand detection is attempted, additional metrics could include hand localization accuracy. For gesture recognition, standard classification accuracy and confusion between gestures would matter. These last metrics will be most significant since they will measure how well our custom model performs.

It would also be useful to test the system under different lighting conditions, distances, and users to see whether it works consistently. This strong human-detection prototype would be a meaningful result and a solid first step toward vision-based human-robot communication.